Performance Analysis of the XGBoost Algorithm for Breast Cancer Diagnosis

Dataset: Breast Cancer Wisconsin (Diagnostic) | Prepared by: Farhan Shayan

1. Overview of the Work

The goal of this project was to classify tumors as benign or malignant using cell measurements taken from biopsy images. The dataset has 569 samples and 30 numeric features, such as radius, perimeter, area, concavity and smoothness of the cell nucleus. Of these samples, 357 are benign and 212 are malignant.

The ID column and an empty trailing column were removed, and the labels were encoded as 0 (benign) and 1 (malignant). The data was split 80/20 with the class ratio preserved, giving 455 samples for training and 114 for testing. The feature scaler was fitted on the training data only and then applied to the test data, so no information from the test set leaked into training.

The final model is an XGBoost classifier with 100 trees, a learning rate of 0.1 and a maximum depth of 1. We chose the depth after trying values from 1 to 10. Depth 1 reached 97% accuracy on the test set while deeper trees reached 96%. The deeper models scored 100% on the training data, which points to memorization rather than real learning.

2. Final Results on the Test Set (114 samples)

Metric	Value	What it means here
Accuracy	97.37%	111 of 114 samples were classified correctly
Precision	100%	Every case flagged as malignant really was malignant
Recall	92.86%	39 of the 42 malignant cases were found
F1-score	96.30%	Harmonic mean of precision and recall
Specificity	100%	All 72 benign cases were identified correctly

The confusion matrix of the model looks like this:

	Predicted: Benign	Predicted: Malignant
Actual: Benign	72	0
Actual: Malignant	3	39

In plain terms, the model never labeled a healthy person as malignant, but it labeled three malignant cases as benign. The features the model relied on most were, in order, concave points_mean, radius_worst, perimeter_worst, concave points_worst and area_worst.

3. Strengths of the Algorithm

- **High accuracy with a simple model.** Trees with a single decision step were enough to reach 97% accuracy. The boundary between the two classes in this data is clear, so a complex model was not needed.
- **Perfect precision.** There were no false alarms. In practice, nobody would be sent into treatment or further tests for a condition they do not have.
- **Overfitting under control.** The gap between training accuracy (99%) and test accuracy (97%) at depth 1 is small. Deeper trees hit 100% on training but did worse on the test set, so choosing the simple model was the right call.
- **Handles non-linear patterns.** The trees are built one after another and each one corrects the errors of the previous ones. This works steadily even when features are correlated, such as radius, perimeter and area.
- **Explainable.** We can say which features mattered most. That is important in a medical setting because a doctor wants to know why the model reached its conclusion.
- **Clean data preparation.** A stratified split, a fixed random seed for reproducibility, and a scaler fitted only on training data make the results more trustworthy.
- **Speed.** Training on data of this size takes a few seconds, and the saved model file is small.

4. Weaknesses and Limitations

- **Recall is not perfect.** Three malignant cases were missed. In cancer screening, missing a real patient is far more dangerous than a false alarm. Improving recall should be the next goal, for example by lowering the decision threshold below 0.5 or by giving the malignant class a higher weight.
- **Small test set.** There are only 42 malignant samples. One more or one fewer mistake moves recall by about 2.4 percentage points, so the numbers should not be read as exact.
- **Depth was chosen using the test set.** We picked the depth by looking at test accuracy, which makes the result slightly optimistic. A separate validation set or cross-validation would be the proper way to do it.
- **Limited tuning.** Only the tree depth was varied. The number of trees, the learning rate and the regularization parameters were not explored and may give better results.
- **Correlated features.** Many features carry duplicate information (radius, perimeter and area say almost the same thing), which makes feature importance harder to interpret.
- **Single data source.** The model was evaluated only on this dataset. For real use it needs to be tested on data from other hospitals and other imaging equipment.

5. Conclusion

> The XGBoost model with depth 1 performed well in this project: 97% accuracy, 100% precision, 93% recall and a 96% F1-score. Still, because of the three missed malignant cases and the small test set, it is for now suitable only as a decision-support tool and should not replace a doctor's judgment. Suggested next steps: cross-validation, tuning the decision threshold to raise recall, and testing on external data.